

Lecture 9: Markov Decision Process and Markov Reward Process

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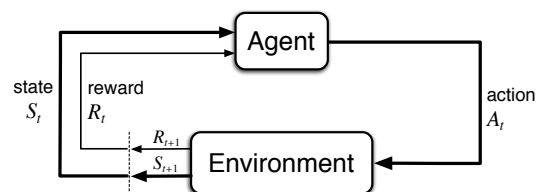
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Finite Markov Decision Processes (MDP)

- ▶ MDPs are a classical formalization of sequential decision making, where actions influence not just immediate rewards, but also subsequent situations, or states, and through those future rewards.
- ▶ Thus MDPs involve delayed reward and the need to trade off immediate and delayed reward.
- ▶ In applications, we need to estimate the value $q_*(a)$ of each action a , in meanwhile we need to estimate the value $q_*(s, a)$ of each action a in each state s , or we estimate the value $v(s)$ of each state given optimal action selections.

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Agent–environment Interaction in a MDP



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Notation of a MDP

- ▶ The sets of states, actions, and rewards are defined as follows
 $\mathcal{S}$, $\mathcal{A}$, and $\mathcal{R}$
- ▶ Given a sequence of discrete time steps
 $t = 0, 1, 2, 3, \dots$
the current state and action are denoted by
 $S_t \in \mathcal{S}$ and $A_t \in \mathcal{A}(s)$
the new reward and state are represented as
 $R_{t+1} \in \mathcal{R} \subset \mathbb{R}$ and S_{t+1}
- ▶ The trajectory of occurrences of state, action, reward over t
 $S_0, A_0, R_1, S_1, A_1, R_2, S_2, A_2, R_3, \dots$

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Markov Property

- ▶ A state S_t is Markov if and only if
$$\mathbb{P}[S_{t+1}|S_t] = \mathbb{P}[S_{t+1}|S_0, S_1, \dots, S_t]$$
- ▶ The state captures all relevant information from the history
- ▶ Once the state is known, the history may be thrown away
- ▶ The current state is a sufficient statistic of the future
 - ▶ i.e. the calculation of S_{t+1} and R_{t+1} will be based on the discrete probability distributions that depend only on the preceding state and action.

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Formulate Dynamics of a MDP

- For particular values of $s' \in \mathcal{S}$ and $r \in \mathcal{R}$, there is a probability of those values occurring at $t+1$, given particular values of the preceding state and action:

$$\mathbb{P}[S_{t+1} = s', R_{t+1} = r | S_t = s, A_t = a] \doteq p(s', r | s, a)$$

- The function $\mathbb{P}$ defines the dynamics of the MDP.
- The dynamics function $\mathbb{P} : \mathcal{S} \times \mathcal{R} \times \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ is an ordinary deterministic function of four arguments.
- p specifies a probability distribution for each choice of s and a , that is,

$$\sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r | s, a) = 1, \text{ for all } s \in \mathcal{S}, a \in \mathcal{A}(s)$$

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State Transition Matrix

- For a Markov state s and successor state s' , the state transition probability is defined by

$$\mathcal{P}_{ss'} = \mathbb{P}[S_{t+1} = s' | S_t = s]$$

- State transition matrix $\mathcal{P}$ defines transition probabilities from all states s to all successor states s' ,

$$\mathcal{P} = \begin{matrix} & \text{to} \\ \text{from} & \begin{bmatrix} \mathcal{P}_{11} & \dots & \mathcal{P}_{1n} \\ \vdots & & \vdots \\ \mathcal{P}_{n1} & \dots & \mathcal{P}_{nn} \end{bmatrix} \end{matrix}$$

where each row of the matrix sums to 1.

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Markov Process

- A Markov process is a memoryless random process, i.e. a sequence of random states $S_1, S_2, \dots$ with the Markov property.

Definition:

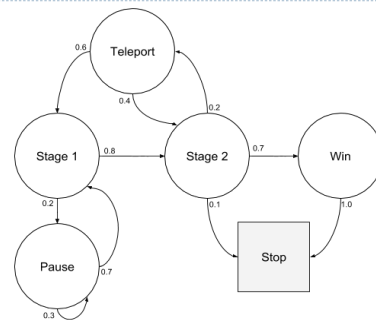
A Markov Process (or Markov Chain) is a tuple $\langle \mathcal{S}, \mathcal{P} \rangle$

- $\mathcal{S}$ is a (finite) set of states
- $\mathcal{P}$ is a state transition probability matrix,
 $\mathcal{P}_{ss'} = \mathbb{P}[S_{t+1} = s' | S_t = s]$

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Example: A Markov Chain



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Sample Episodes

- A sample episode is to go through the chain and end up at the terminal state.
- An example sample episode would be to go from Stage1 to Stage2 to Win to Stop. Below is a representation of a few sample episodes:

- $S_1 S_2 \text{Win Stop}$
- $S_1 S_2 \text{Teleport } S_2 \text{Win Stop}$
- $S_1 \text{Pause } S_1 S_2 \text{Win Stop}$

- These Markov Chains have the following Transition Probability Matrix:

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Example of the Transition Matrix

$$\mathcal{P} = \begin{matrix} & \begin{matrix} \text{Stage1} & \text{Stage2} & \text{Teleport} & \text{Pause} & \text{Win} & \text{Stop} \end{matrix} \\ \begin{matrix} \text{Stage1} \\ \text{Stage2} \\ \text{Teleport} \\ \text{Pause} \\ \text{Win} \\ \text{Stop} \end{matrix} & \begin{bmatrix} 0.6 & 0.8 & 0.2 & 0.2 & 0.7 & 0.1 \\ 0.7 & 0.4 & 0.2 & 0.3 & 1.0 & 1.0 \end{bmatrix} \end{matrix}$$

For each of the states the sum of the transition probabilities for that state equals 1.

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Markov Reward Process (MRP)

- ▶ A Markov reward process is a Markov chain with values.

Definition:

A Markov Reward Process is a tuple $\langle \mathcal{S}, \mathcal{P}, \mathcal{R}, \gamma \rangle$

- $\mathcal{S}$ is a finite set of states
- $\mathcal{P}$ is a state transition probability matrix,
 $\mathcal{P}_{ss'} = \mathbb{P}[S_{t+1} = s' \mid S_t = s]$
- $\mathcal{R}$ is a reward function, $\mathcal{R}_s = \mathbb{E}[R_{t+1} \mid S_t = s]$
- γ is a discount factor, $\gamma \in [0, 1]$

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Return

- ▶ The goal of Reinforcement learning is to maximize the cumulative reward
 - ▶ all the rewards agent receives from the environment
 - ▶ instead of, the reward agent receives from the current state
- ▶ This total sum of reward the agent receives from the environment is called **returns**.

$$G_t \doteq R_{t+1} + R_{t+2} + R_{t+3} + \dots + R_T$$

- ▶ R_{t+1} is the reward received by the agent at time step $t=0$ while moving from one state to another.
- ▶ Similarly, R_T is the reward received by the agent at the final time step $t=T$ when mapping it to final state.

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Return (cont'd)

- ▶ In other word, the goal is to maximise the **return**. The return G_t is the total discount reward from time-step t .

$$G_t = R_{t+1} + \gamma R_{t+2} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

- ▶ The discount $\gamma \in [0, 1]$ is the present value of future rewards
- ▶ The value of receiving reward R after $k + 1$ time-steps is $\gamma^k R$.
- ▶ This values immediate reward above delayed reward.
 - ▶ γ close to 0 leads to "myopic" evaluation
 - ▶ γ close to 1 leads to "far-sighted" evaluation

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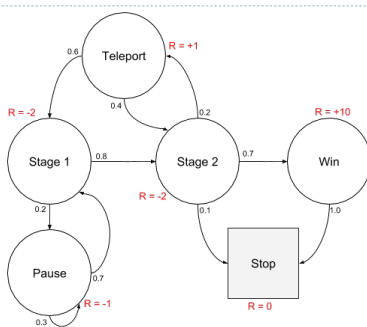
Why discount?

- ▶ Most Markov reward and decision processes are discounted with following reasons:
 - ▶ Mathematically convenient to discount rewards
 - ▶ Avoids infinite returns in cyclic Markov processes
 - ▶ Uncertainty about the future may not be fully represented
 - ▶ If the reward is financial, immediate rewards may earn more interest than delayed rewards
 - ▶ Animal/human behaviour shows preference for immediate reward
 - ▶ It is sometimes possible to use undiscounted Markov reward processes (i.e. $\gamma = 1$), e.g. if all sequences terminate.

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Example for Calculating Returns



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Example: Return Values

- ▶ Sample returns for the Teleport MRP:
- ▶ Starting from $S_1 = \text{Stage I}$ with $\gamma = 0.5$

$$G_1 = R_2 + \gamma R_3 + \dots + \gamma^{T-2} R_T$$

Sample episodes

$S_1 S_2 \text{Win Stop}$	$G_1 = -2 - 2 * 0.5 + 10 * 0.25 = -0.5$
$S_1 S_2 \text{Teleport } S_2 \text{Win Stop}$	$G_1 = -2 - 1 + 0.25 - 0.25 + 0.625 = -2.375$
$S_1 \text{Pause } S_1 S_2 \text{Win Stop}$	$G_1 = -2 - 0.5 - 0.5 - 0.25 + 0.625 = -2.625$

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Policy Function

- ▶ Value Function determines how good it is for the agent to be in a particular state.
- ▶ To determine how good it will be to be in a particular state it must depend on some actions that it will take.
 - ▶ A policy defines what actions to perform in a particular state s .
- ▶ A policy is a simple function, that defines a probability distribution over Actions ($a \in A$) for each of States ($s \in S$).
- ▶ If an agent at time t follows a policy π then $\pi(a | s)$ is the probability that the agent with taking action a at a particular time step t .

$$\pi(a|s) = \mathbb{P}[A_t = a | S_t = s]$$

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Value Function

- ▶ How do we find a value of a state. The value of state s , when the agent is following a policy π which is denoted by $v(s)$ is the expected return
 - ▶ starting from s and following a policy π for the next states until we reach the terminal state.
- ▶ We can formulate a value function as:

$$v(s) = \mathbb{E}[G_t | S_t = s]$$

- ▶ This equation gives us the expected returns starting from the state s and going to successor states thereafter, with the policy π .
- ▶ It should be noted that the value of the terminal state is zero.

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Bellman Equation

- ▶ The value function can be decomposed into two parts:
 - ▶ immediate reward R_{t+1}
 - ▶ discounted value of successor state $\gamma v(S_{t+1})$

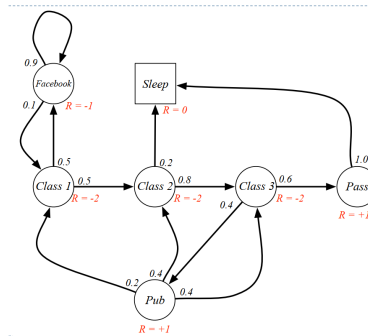
$$\begin{aligned} v(s) &= \mathbb{E}[G_t | S_t = s] \\ &= \mathbb{E}[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | S_t = s] \\ &= \mathbb{E}[R_{t+1} + \gamma (R_{t+2} + \gamma R_{t+3} + \dots) | S_t = s] \\ &= \mathbb{E}[R_{t+1} + \gamma G_{t+1} | S_t = s] \\ &= \mathbb{E}[R_{t+1} + \gamma v(S_{t+1}) | S_t = s] \end{aligned}$$

- ▶ What the equation means is that we only need the next state to calculate the total value of a state

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Example: Student Study



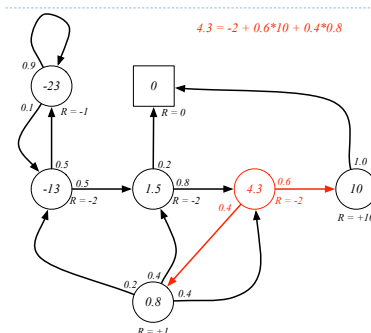
Markov process student activities: from state Class 1 until Sleep which is the final state.

- The numbers beside each circle represent the transition probabilities.
- Red numbers represent rewards

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State-Value with Bellman Equation



- Gamma equals to 1
- Calculate the value in Class 3 (red circle) with the equation
- by summing the immediate reward (-2) with the expected value of two next states.

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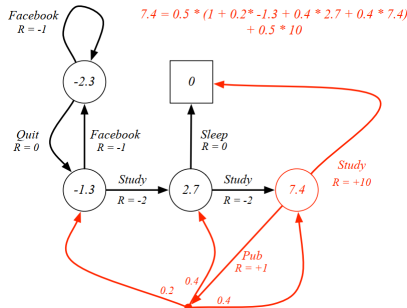
State-Value with Bellman Equation

- ▶ To make things clearer, we can look at the Student MDP with gamma 0.1
- ▶ Let's say that in Class 3 (red circle), a student has a 50:50 policy.
- ▶ It means that the student has a 50% chance to study (Study) or go to a pub (Pub).
- ▶ We can calculate the state value by summing each expected return that follows each action.

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State-Value with Bellman Equation



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State-Value with Bellman Equation

- ▶ The expected value from Study can be calculated by multiplying the action probability and the expected value of the next state (0.5×10).
- ▶ In contrast, Pub action has multiple branches that lead to different states.
- ▶ Therefore, we can calculate the expected value from Pub by multiplying the action probability (0.5) by the action-value.
- ▶ The action-value can be calculated by summing the immediate reward with expected values from all of the possible states.
- ▶ It can be calculated by $1 + 0.2 \times -1.3 + 0.4 \times 2.7 + 0.4 \times 7.4$.

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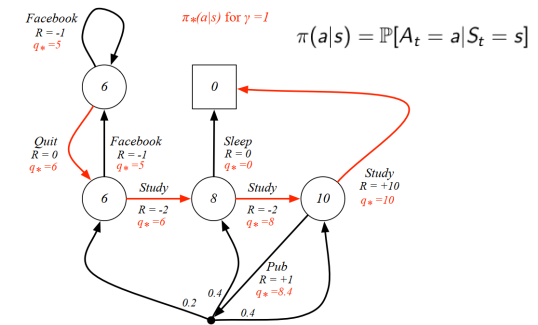
Example Policy Function

- ▶ The goal of an agent is to maximize its value. So, we have to find optimal value functions which result in maximum value on the previous example
- ▶ we show the calculation of value by summing all of the expected values from all possible actions.
- ▶ Now, we only care about the action that gives the maximum value.
- ▶ After having known the optimal value functions, we have the optimal policy and the MDP is solved.

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Optimal Value Function for Student MDP



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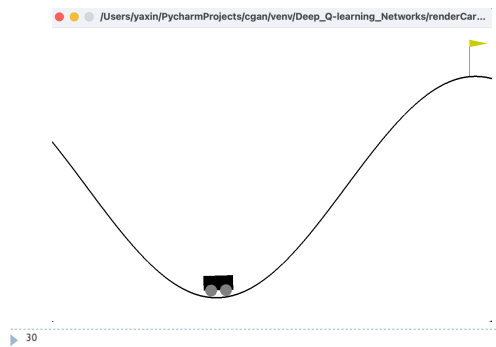
Optimal Policy

- ▶ The red arcs are the optimal policy which means that if our agent follows this path it will yield maximum reward from this MDP.
- ▶ Also, by seeing the q^* values for each state we can say the actions our agent will take that yields maximum reward.
- ▶ So, optimal policy always takes action with higher q^* value (State-Action Value Function).
- ▶ For example, in the state with value 8, there is q^* with value 0 and 8. Our agent chooses the one with greater q^* value i.e. 8.

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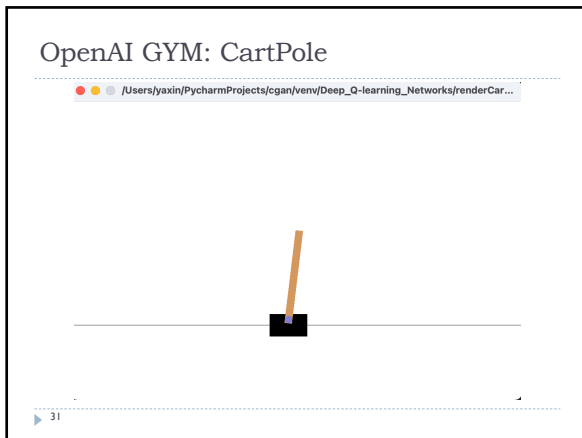
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OpenAI GYM: MountainCar



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